Completed

First-Come,First-Served CPU Scheduling Algorithm

Write a C program to implement the First-Come, First-Served (FCFS) CPU Scheduling Algorithm. The program should take the number of processes, their Arrival Times (AT), and Burst Times (BT) as input. Calculate and display the Completion Time (CT), Turnaround Time (TAT), and Waiting Time (WT) for each process, along with the Average Turnaround Time and Average Waiting Time.

To evaluate the performance of the algorithm, we use the following formulas:

- **Completion Time (CT):** The time at which a process finishes execution.
- **Turnaround Time (TAT):** Total time taken from arrival to completion.
- $TAT = CT - AT$
- **Waiting Time (WT):** The time a process spends waiting in the ready queue.
- $WT = TAT - BT$

Input Format

1. The first input line contains an integer representing the number of processes.
2. The next lines contain two integers for each process, where the first

Select Language : C (GCC 9.2.0)

```c
#include <stdio.h>
int main()
{
  int n;
  scanf("%d",&n);
  printf("Enter number of processes: \n");
  int AT[n],BT[n],CT[n],TAT[n],WT[n];
  for(int i=0;i<n;i++)
  {
    printf("Enter Arrival Time and Burst Time for P%d: \n",i+1);
    scanf("%d %d",&AT[i],&BT[i]);
  }
  CT[0]=AT[0]+BT[0];
  for(int i=1;i<n;i++)
  {
    if(CT[i-1]<AT[i])
    {
      CT[i]=AT[i]+BT[i];
    }
    else
    {
      CT[i]=CT[i-1]+BT[i];
    }
  }
  float totalTAT=0,totalWT=0;
```

Test Cases 2

Run Sample Cases

Run All Cases

Completed

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```c
  for(int i=1;i<n;i++)
  {
    if(CT[i-1]<AT[i])
    {
      CT[i]=AT[i]+BT[i];
    }
    else
    {
      CT[i]=CT[i-1]+BT[i];
    }
  }
  float totalTAT=0,totalWT=0;
  for(int i=0;i<n;i++)
  {
    TAT[i]=CT[i]-AT[i];
    WT[i]=TAT[i]-BT[i];
    totalTAT=totalTAT+TAT[i];
    totalWT=totalWT+WT[i];
  }
  float avgTAT=totalTAT/n;
  float avgWT=totalWT/n;
  printf("\nAverage Turnaround Time: %.2f\n",avgTAT);
  printf("Average Waiting Time: %.2f\n",avgWT);
  return 0;
}
```

Test Cases 2

Run Sample Cases

Run All Cases